

EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 22- 302 - 2

OUTDOOR LED LUMINARIES

DATE :9-12- 2025

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**TECHNICAL SPECIFICATION
FOR
OUTDOOR LED LUMINARIES**

Issue: DEC -2025/ Rev-2

سيتم بدء العمل بالمواصفة من ٢٠٢٦/٧/١

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EEHC DISTRIBUTION MATERIALS SPECIFICATION**EDMS 22- 302 - 2****OUTDOOR LED LUMINARIES****DATE :9-12- 2025****1 - Scope:**

This specification covers the minimum technical requirements for design, engineering, Manufacturing, testing and Packing for outdoor LED luminaires . The required luminaires are to be installed in LV networks of the Egyptian Electricity Distribution companies. The outdoor LED luminaires includes Street lighting, lighting under bridges , lighting the Tunnels, railway lighting, clubs lighting etc .

2 - Applicable standards:

Unless otherwise specified in this specification, The outdoor LED Lighting Luminaires Should be designed, manufactured and tested according to the requirements of this specification, taking into consideration the recommendations and requirements of the latest editions of the following standards:

IEC 60598-2-3 part2	luminaires for road and street lighting
IEC 61347-2-13	DC or AC supplied electronic control gear for led modules
IEC 61000-3-2	limitation of harmonic distortion
EN 61000-3-3	Limitation of voltage fluctuation and flicker
IEC 61347-1	General safety requirement
IEC 62612	LED lamps for general lighting services with supply voltages > 50 V
IEC 62717	LED modules for general lighting - Performance Requirements
IEC 62722-2-1	Luminaire performance - Part 2-1: Particular requirements for LED Luminaires
IEC 62722	Estimation of luminous flux maintenance
EN 60529	Degrees of protection provided by enclosures (IP Code)
IEC 60068	environmental testing for led chip
EN 60598-1	Luminaires - Part 1: General requirements and tests
IEC 62504	Light emitting diode (LED) products and related equipment
ANSI C78.337	The chromaticity of solid state lighting product For electric lamps
IEC/ EN 62262	Degree of protection against external mechanical Impact (IK code)
IECQ OD 3803:2017	LED driver reliability testing
IEC 62384	DC or AC supplied electronic control gear for LED modules

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3 – Environmental Conditions:

Ambient Temperature (°C)	-5 to 50°C
Humidity	10% -90%.

4 – Technical specification:

The outdoor LED luminaires consist of three main parts : The body ,The driver and the LED chips . The required LED luminaires must comply the following general specifications

Rated power (W)	50 up to 200 according to EDC demand
Beam angle	$\geq 120^\circ \times 60^\circ$
Installation Diameter	Pole or Arm diameter of 60MM
Efficacy	≥ 120 LM/W
Life Time	50000 H (L 80)

4.1 The body

Material of manufacture	High Pressure Die Cast Aluminum
IP	66
Insulation class	1
Glass	Glass is thermally resistant and has a mechanical degree against breakage (IK08)
Protection	Equipped With Heat Sink to protect the led from overheating
Painting	electrostatic powder painted ,with thickness not less than 60 microns (or marine salt painting according to EDC requirement)

- The glass should be completely sealed on the body and does not allow air, dust and Water enter through it
- The fixture arm should be made of the same material as the body, so that it is not rust and weatherproof

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IP	65
PF	≥ 95
Frequency (HZ)	50
Life Time	≥ 50000 H
THD	Not more than 20 % for Current and Not more than 5 % for Voltage
Voltage range	100 to 265 V
Over Temp Protection	Equipped with protection against over temp $> 80^{\circ}\text{C}$
Over voltage Protection	Equipped with protection against over voltage > 265 v
Operating Temp	-5 to $+70^{\circ}\text{C}$
SPD (surge protection device)	Not less than 6 KV
Operating Current	The supply and output current according to IEC 62384
Insulation resistance	$> 4\text{ M}\Omega$
efficiency	$\geq 90\%$
Dimming	Equipped with Auto Dimming feature (as option)

4.3 The LED chips

CCT (Correlated Colour Temperature)	4000 K to 6400 K according to EDC demand
CRI (Colour Rendering Index)	≥ 80
Life Time	50000 H
Lumen Maintenance code (LM)	L80
Operating Temperature	-5 to 105°C
SDCM (standard deviation color matching)	step 5 Macadam ellipse

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5- Inspection and Testing :

5-1 Routine tests

- Test of marking
- Insulation resistance test
- Electrical and optical test
- IP test
- Mechanical strength test
 -EDC preserves its right to witness the routine test of the luminaire and its components.
 - In the absence of an approved laboratory at suppliers factory or company a random samples shall be tested with any approved laboratory EDC specify at the expense of the supplier
 - Unless otherwise specified or approved in writing byEDC, all lamps should be Tested In accordance with all relevant latest applicable mentioned standards and test certificate should be provided

5-2 LED luminaries tests certificates

The supplier should provide the following recently updated certificates:

- **LM 80 test report (on LED chip level)**
- **LM 79 test report(on system level) or what is required of it in one of the accredited Egyptian laboratories, which includes the following tests**
 - Total Luminous Flux (light output)
 - Luminous Intensity Distribution
 - Electrical Power Characteristics (wattage, voltage...)
 - Luminous Efficacy (calculated)
 - Color Characteristics (CRI, CCT...)
 - SDCM (Macadam ellipse) report
- **IP&IK test report**
- **Fragmentation test for glass**

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6-Marking

The marking should be according to the following table

	Items	Place			
		Carton	Product *	Datasheet	IEC standard
Marking	Mark of origin (trademark, manufactures name)	√	√	√	60598-1
	Rated Voltage (V)	√	√	√	
	Rated max. Ambient temp. (C)	√	√	√	
	IP Number	√	√	√	
	Rated current (A)			√	
	Rated Frequency (Hz)	√	√	√	
	Beam angle	√	√	√	
	Rated input power (w)	√	√	√	627722-2-1
	Rated luminous flux (lm)	√	√	√	
	Life Time (h)	√	√	√	
	Rated abrupt failure value (%)			√	
	Lumen maintenance code			√	
	Correlated Colour Temperature(CCT)	√	√	√	
	Colour rendering index (CRI)	√	√	√	
	Rated efficacy (Lm/w)		√	√	
	Weight including control gear			√	60598-2-3
	Overall dimensions			√	

* The marks on the product should not be erasable

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7-Packing

The luminaire boxes should be packed in suitable cartoons (cardboard box) to withstand transporting, handling and storing

8-Warranty , Catalog and sample

- 1- The manufacture / tenderer should guarantee the luminaire against all defects for (3) years starting from the date of delivery .
- 2- The tenderer should fill in the attached guarantee technical data table.
- 3- The tenderer should attach with his offer the original catalogue for all components used with catalogue numbers marked clearly for the offered luminaire (soft or hard copies) .
- 4- The tenderer should submit with his offer sample of each item for testing and the offer will be rejected without offer a sample.

9-Wiring and Connections

- The internal connections of the luminaire are made of thermoplastic flexible copper wire with a cross-section area not less than 1 mm^2 and the wires bear a temperature not less Than 105°C .
- The internal connections between the driver and the chips by sockets

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10-Guarantee table

	EDC Requirement	Supplier Proposal
Country of origin		
Manufacturer name		
Year of manufacturing		
Brand name or type		
Standard specification		
Rated power		
Voltage range	100 to 265 V	
Beam angle	$\geq 120^\circ \times 60^\circ$	
IP(the body)	66	
PF	≥ 95	
Frequency (HZ)	50	
Life Time of luminaire	50000 H (L80)	
Operating Current	according to IEC 62384	
Operating Temp for drive	-5 to +70 °C	
THD i% ,THD v %	20% , 5 %	
Efficacy of Luminaire	≥ 120 LM/W	
CCT (Correlated Colour Temperature)	4000 K to 6400 K	
CRI (Colour Rendering Index)	≥ 80	
SPD (surge protection device)	Not less than 6 KV	
Body and housing material	High Pressure Die Cast Aluminum	
mechanical degree against breakage (IK)	(IK08)	
Painting	electrostatic powder, with thickness not less than 60 microns	
efficiency Driver	≥ 90 %	
Over Temp Protection	> 80 °C	
Over voltage Protection	> 265 v	